

GWQ: Group-Wise Quantization Framework for Neural Networks

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Abstract

As the most commonly used quantization techniques for deep neural networks, the int-only quantization methods use scale factor to linearly approximate the weights or activation of each layer. However, when passing activation data between layers, such int-only quantization methods require extra Scale Factor Conversion (SFC) operations, resulting in computational overhead. In this paper, we propose a Group-Wise Quantization framework, called GWQ, to reduce computational consumption during the activation data pass process by allowing multiple layers share one scale factor in SFC operations. Specifically, in the GWQ framework, we propose two algorithms for network layers grouping and model training. For the grouping of network layers, we propose a grouping algorithm based on the similarity of data numerical distribution. Then, the network layers divided into the same group will be quantified using the same common scale factor to reduce the computational consumption. Considering the additional performance loss caused by sharing scale factors among multiple layers, we propose a training algorithm to optimize these shared scale factors and model parameters, by designing a learnable power-of-two scaling parameter for each layer. Extensive experiments demonstrate that the proposed GWQ framework is able to effectively reduce the computational burden during inference, while maintaining model performance with negligible impact.

Keywords: Int-Only Quantization; Scale Factor Conversion; Quantization-Aware-Training; Deep Neural Networks; Grouping-Wise Algorithm

1. Introduction

Benefiting from the rapid development of computing power and storage technology in recent years, deep learning models with a vast number of parameters are designed to handle increasingly complex tasks. However, running these models often consumes significant computational resources. Those expensive costs hinder the deployment of deep learning in industry, agriculture, and vehicle areas. How to design high-efficiency deep learning

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